الموضوع: Back Propagation التاريخ: / /

1- Net h_1 = $i_1 * w_1 + i_2 * w_2 + b_1$
= ,05 * ,15 + ,1 * ,2 + ,35
= ,3775

output

2- O h_1 = $\sigma(,3775) = \frac{1}{1+e^{-,3775}}$
= ,59327

3- Net h_2 = $i_1 * w_3 + i_2 * w_4 + b_1$
= ,05 * ,25 + ,1 * ,3 + ,35
= ,3925

4- O h_2 = $\sigma(,3925) = \frac{1}{1+e^{-,3925}}$
= ,59688

5- Net o_1 = $oh_1 * w_5 + oh_2 * w_6 + b_2$
= ,59327 * ,4 + ,59688 * ,45 + ,6
= 1,1059

6- O o_1 = $\sigma(1,1059) = \frac{1}{1+e^{-1,1059}}$
= ,75136

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7- $Net\ O_2 = Oh_1 * W_7 + Oh_2 * W_8 + b_2$

$= ,59327 * ,5 + ,59688 * ,55 + ,6$

$= 1,2249$

8- $O\ O_2 = \sigma(1,2249) = \frac{1}{1+e^{-1,2249}}$

$= ,77293$

9- $MSE = \frac{1}{2}(t_1 - O_1)^2 + \frac{1}{2}(t_1 - O_2)^2$

$= \frac{1}{2}(,01 - ,75136)^2 + \frac{1}{2}(,99 - ,77293)^2$

$= ,29837$

The Backward Pass

1- $\delta O_1 = O_1 * (1 - O_1) * (t_1 - O_1) =$

$= ,75136 * (1 - ,75136) * (,01 - ,75136)$

$= -,1385$

2- $\delta O_2 = O_2 * (1 - O_2) * (T_2 - O_2)$

$= ,77293 * (1 - ,77293) * (,99 - ,77293)$

$= ,0381$

الموضوع : update weights التاريخ : / /

$* W_{new} = W_{old} + \eta * S * X_i$

$\eta = ,5$

$W_5 = ,44,5 * -,1385 * ,59327 = ,35892$

$W_6 = ,484,5 * -,1385 * ,59688 = ,40866$

$W_7 = ,5 * ,5 * ,0381 * ,59327 = ,51130$

$W_8 = ,55 * ,5 * ,0381 * ,59688 = ,56137$

S for hidden layer

$* S = O * (1 - O) * S * W$

$Sh_1 = ,59327(1 - ,59327) * (-,1385 * ,4 + ,0381 * ,5)$
$= -,00877$

$Sh_2 = ,59688(1 - ,59688) * (-1,385 * ,45 + ,0381 * ,55)$
$= -,00995$

update weights

$W_1 = ,25 + ,5 * -,00877 * ,05 = ,14978$

$W_2 = ,2 + ,5 * -,00877 * ,1 = ,19956$

الموضوع : التاريخ : / /

$W_3 = ,25 * ,5 * -,00995 * ,05 = ,24975$

$W_4 = ,3 * ,5 * -,00995 * ,1 = ,2995$

Update Bias $b_{new} = b_{old} + \eta * \delta$

$b_1 = ,35 + ,5 * (-,00995 + -,00877) = ,34064$

$b_2 = ,6 + ,5 * (-,13884 + ,0381) = ,5498$

W_1	W_2	W_3	W_4	W_5	W_6	W_7	W_8	B_1	B_2
,3497	,2995	,24975	,2995	,3589	,40866	,5113	,56137	,3406	,5498